

Test 10 — Question Paper

38 items. Suggested time: 95 minutes (about 2 minutes 30 seconds per item).
A four-function calculator may be used **only** on items marked *calculator*.
The Grade 6 B.E.S.T. Mathematics Reference Sheet is on the last page.
Aligned to the FAST Grade 6 Mathematics test design: 33–42% Number Sense and Operations, 25–36% Algebraic Reasoning, 25–36% Geometric Reasoning, Data Analysis, and Probability.

1. Match each power with its value.

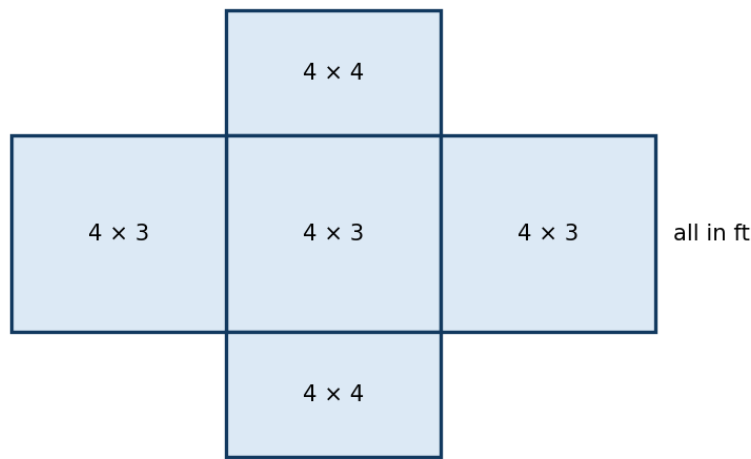
	8	30	100	1000
10^2	☐	☐	☐	☐
10^3	☐	☐	☐	☐
2^3	☐	☐	☐	☐

Show your work

calculator · matching table

2. The net of a right rectangular prism is shown.

What is the surface area of the prism, in square feet?



- A. 32
- B. 48
- C. 80
- D. 40

Show your work

multiple choice

3. Match each number with the equivalent percent.

	25%	50%	60%	67%
1/2	■	■	■	■
0.6	■	■	■	■
1/4	■	■	■	■

Show your work

calculator · matching table

4. Which number is located between -2 and -1 on a number line?

- A. $-1 \frac{3}{4}$
- B. 0
- C. $-\frac{3}{4}$
- D. $-2 \frac{1}{2}$

Show your work

calculator · multiple choice

5. A sixth-grade class at Palmetto Middle has 25 students who ride the bus and 15 students who walk.

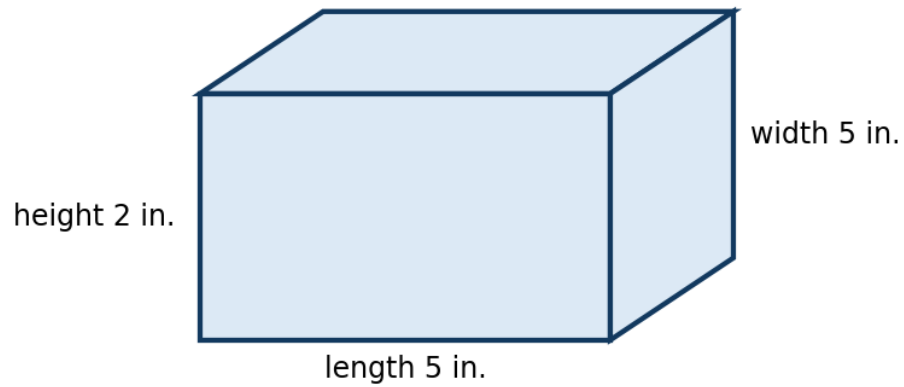
What is the ratio of students who ride the bus to students who walk, in simplest form?

- A. 3:5
- B. 8:3
- C. 5:8
- D. 5:3

Show your work

multiple choice

6. What is the volume of the right rectangular prism shown, in cubic inches?



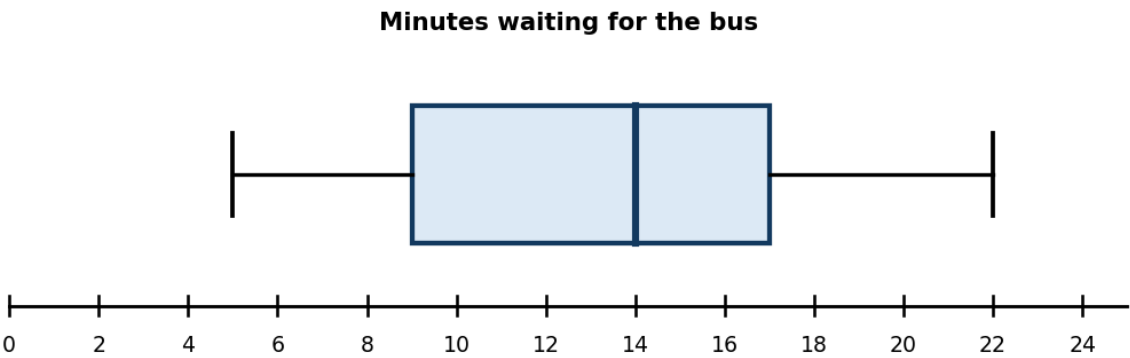
- A. 90
- B. 25
- C. 12
- D. 50

Show your work

multiple choice

7. The box plot shows a set of data.

What is the median of the data?

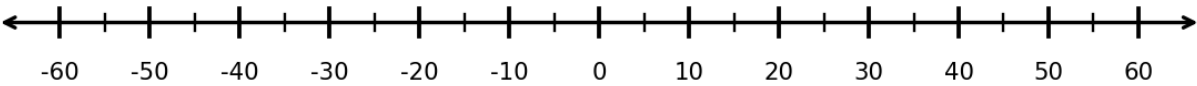


- A. 9
- B. 16
- C. 17
- D. 14

Show your work

multiple choice

8. Plot the value of $-40 + 30$ on the number line.



Mark your answer on the diagram above.

Show your work

calculator · graphing (grid)

9. Use the drop-down menus to complete the statements about 80 students.

80% of 80 is students, and 50% of 80 is students.

Blank (1) choices: 0 | 64 | 80 | 128

Blank (2) choices: 20 | 40 | 80 | 160

Show your work

editing task choice

10. The table shows equivalent ratios of buses to vans.

Use the drop-down menus to complete the table: the first ? is and the second ? is .

Buses	Vans
3	5
?	10
9	?

Blank (1) choices: 3 | 5 | 6 | 10

Blank (2) choices: 5 | 8 | 9 | 15

Show your work

editing task choice

11. What is the value of $3x^2 - 12$ when $x = 3$?

- A. 6
- B. 69
- C. -9
- D. 15

Show your work

multiple choice

12. A carry-on bag must weigh no more than 48 pounds.

Let x represent the weight in pounds. Which inequality represents this situation?

- A. $x < 48$
- B. $x > 48$
- C. $x \leq 48$
- D. $x \geq 48$

Show your work

multiple choice

13. What is the solution to the equation $2x = 10$?

- A. 5
- B. 8
- C. 20
- D. 10

Show your work

multiple choice

14. Which expression represents a number x decreased by 9?

- A. $x + 9$
- B. $9 - x$
- C. $x - 9$
- D. $x/9$

Show your work

multiple choice

15. Point M is at $(-4, -5)$ and point N is at $(1, -5)$.

What is the distance, in units, between point M and point N?

- A. 3
- B. 9
- C. 5
- D. -3

Show your work

multiple choice

16. Buses on one route leave Earlington Heights every 12 minutes and buses on another route leave every 18 minutes. Both leave at 9:00 a.m.

In how many minutes will they next leave at the same time?

- A. 216
- B. 6
- C. 30
- D. 36

Show your work

calculator · multiple choice

17. What is the value of $2(3 + 5)$?

- A. 11
- B. 30
- C. 16
- D. 10

Show your work

calculator · multiple choice

18. A point is graphed at $-3\frac{3}{4}$ on a number line.

How many units is the point from 0?

- A. $1\frac{7}{8}$
- B. $-3\frac{3}{4}$
- C. $4\frac{3}{4}$
- D. $3\frac{3}{4}$

Show your work

calculator · multiple choice

19. What value of x makes the equation $x + 1.04 = 8.82$ true?

- A. 7.78
- B. 8.82
- C. 1.04
- D. 9.86

Show your work

multiple choice

20. The table shows the number of minutes six students spent on the bus.

What is the mean number of minutes?

Minutes	#1	#2	#3	#4	#5	#6
Minutes	13	15	25	26	29	30

- A. 138
- B. 23
- C. 17
- D. 26

Show your work

multiple choice

21. Which question is a statistical question?
- A. How many minutes long is my bus ride to school?
 - B. How many minutes did each sixth grader spend on the bus this morning?
 - C. How many books did I read last month?
 - D. How tall is the tallest student in my class?

Show your work

multiple choice

22. Match each measurement with its equivalent value. Use the Grade 6 reference sheet.

	12	32	50	500
2 pounds, in ounces	■	■	■	■
3 gallons, in quarts	■	■	■	■
5 meters, in centimeters	■	■	■	■

Show your work

matching table

23. Diego pays with \$20 for 4 items that cost \$3.38 each.

How much change should Diego receive?

- A. \$16.62
- B. \$13.52
- C. \$6.48
- D. \$9.86

Show your work

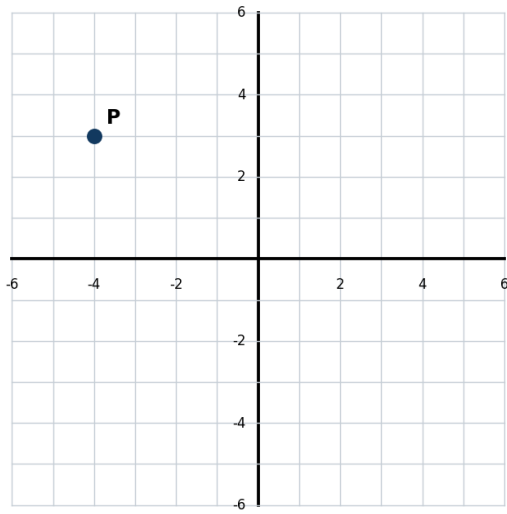
calculator · multiple choice

24. What is the value of $(\frac{2}{5})^2$?
- A. $\frac{2}{5}$
 - B. $\frac{4}{5}$
 - C. $\frac{4}{25}$
 - D. $\frac{2}{25}$

Show your work

calculator · multiple choice

25. What are the coordinates of point P?



- A. (3, -4)
- B. (4, 3)
- C. (-4, 3)
- D. (-4, -3)

Show your work

multiple choice

26. Which expression is equivalent to $4(x + 10)$?

- A. $x + 40$
- B. $4x + 10$
- C. $4x + 40$
- D. $4x + 14$

Show your work

multiple choice

27. A data set is shown.

9, 12, 17, 19, 23

After 32 is added to the data set, match each measure with its new value.

	18	$18\frac{2}{3}$	23	26
Mean	☐	☐	☐	☐
Median	☐	☐	☐	☐
Range	☐	☐	☐	☐

Show your work

matching table

28. What is the opposite of $-5\frac{3}{4}$?

- A. $-11\frac{1}{2}$
- B. $5\frac{3}{4}$
- C. $-5\frac{3}{4}$
- D. $2\frac{7}{8}$

Show your work

calculator · multiple choice

29. A data set is shown.

1, 1, 1, 1, 2, 4, 5, 6, 8, 19, 22

How many values belong in the interval 13-18 of a histogram? Enter your answer.

Answer: _____

Show your work

equation editor

30. Use the drop-down menus to complete the equations.

$$(-6)(-8) = \text{[[1]]} \text{ and } (48) \div (-8) = \text{[[2]]}$$

Blank (1) choices: -48 | -14 | 2 | 48

Blank (2) choices: -8 | -6 | 6 | 48

Show your work

calculator · editing task choice

31. What is $0.625 \div 0.125$?

- A. 0.5
- B. 5
- C. 50
- D. 0.078125

Show your work

calculator · multiple choice

32. Which value of x makes the equation $8x + 20 = 44$ true?

- A. 3
- B. 4
- C. 2
- D. 24

Show your work

multiple choice

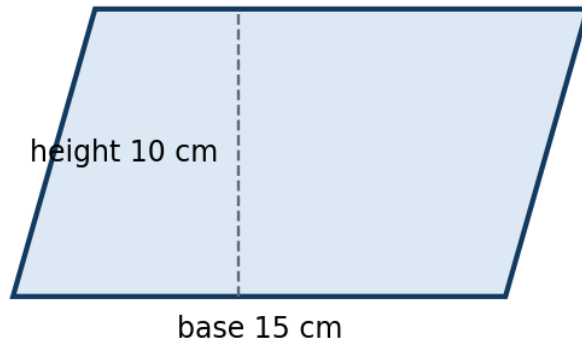
33. What is the solution to the equation $x + 11 = 38$?

- A. 38
- B. 27
- C. 49
- D. 11

Show your work

multiple choice

34. What is the area of the parallelogram shown, in square centimeters?



- A. 150
- B. 25
- C. 75
- D. 50

Show your work

multiple choice

35. Select **all** the prime factors of 126.

Circle every correct choice:

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 7
- ☐ 126

Show your work

calculator · selectable hot text

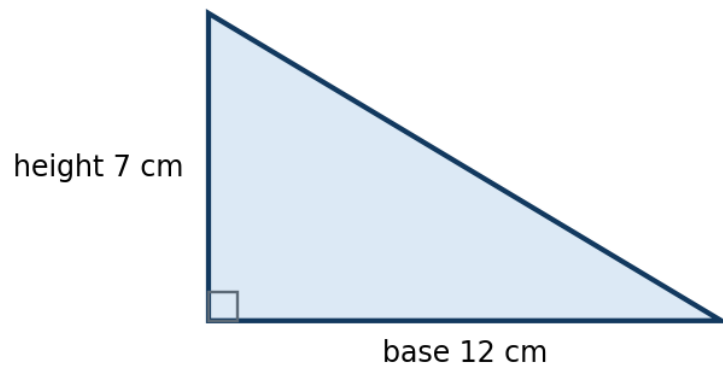
36. What is the value of $|-16| - |10|$?

- A. 8
- B. -6
- C. 26
- D. 6

Show your work

calculator · multiple choice

37. What is the area of the right triangle shown, in square centimeters?



- A. 42
- B. 84
- C. 19
- D. $9 \frac{1}{2}$

Show your work

multiple choice

38. What is $1 \frac{3}{5} \div \frac{1}{2}$?

- A. $\frac{4}{5}$
- B. $3 \frac{1}{5}$
- C. $\frac{5}{16}$
- D. $2 \frac{1}{10}$

Show your work

calculator · multiple choice

Grade 6 Mathematics Reference Sheet

Customary Conversions

1 foot = 12 inches	1 cup = 8 fluid ounces
1 yard = 3 feet	1 pint = 2 cups
1 mile = 5,280 feet	1 quart = 2 pints
1 mile = 1,760 yards	1 gallon = 4 quarts
1 pound = 16 ounces	1 ton = 2,000 pounds

Metric Conversions

1 meter = 100 centimeters	1 liter = 1,000 milliliters
1 meter = 1,000 millimeters	1 gram = 1,000 milligrams
1 kilometer = 1,000 meters	1 kilogram = 1,000 grams

Time Conversions

1 minute = 60 seconds	1 week = 7 days
1 hour = 60 minutes	1 year = 365 days
1 day = 24 hours	1 year = 52 weeks

Formula

Rectangular prism	$V = lwh$ or $V = Bh$
l = length, w = width, h = height	B = area of base, V = volume